

MATH 252 (CRN: 23526), WINTER 2025: CALCULUS II

Lectures. MTWF 10am-10:50am at Peterson 105

Instructor. Dr. Xiaojing Chen–Murphy

- Email: xchenm@uoregon.edu
- Office Hours: Mondays and Fridays 1-2pm, Tuesdays 12-12:50 in Tykeson 333 (Common Study Area)

Course Material.

- *Textbook and Materials:* OpenSTAX Calculus Volume II. An electronic edition of this text is available for free at <http://openstax.org/details/books/calculus-volume-2>.
- *Calculators:* Calculators are not allowed in any of the quizzes or exams.
- *Prerequisite:* C– or better in Math 251.
- *Course Content:* This course begins with a brief review on antiderivatives in Section 4.10 of OpenSTAX Calculus Volume I (<https://openstax.org/details/books/calculus-volume-1>), and then covers most of Chapters 1–4 in the OpenSTAX Calculus Volume II. Topics covered in this course are listed as follows: antiderivatives and notion of integrals; the Fundamental Theorem of Calculus, and various techniques for calculating integrals; additional techniques of integration, including integration by parts, trigonometric integrals and trigonometric substitution, and improper integrals; applications of the integral such as computing areas, and volumes, and applications in physics; and introduction to ordinary differential equations.

Homework, Quizzes and Exams.

- **WebWork Homework (15%).** WebWork homework assignments are due every Tuesday and most Fridays, and will be closed two days after their due dates.
 - The period after the homework due date and before it is closing date is referred as “Reduced Scoring Period” on WebWork. Homework problems that are completed during reduced scoring period will result in a 20% penalty. One lowest WebWork homework will be dropped.
- **Quizzes (8%).** There will be weekly quizzes on Wednesdays.
 - Quizzes are about 10-minute long. Quiz may take place at any time during the 50-minute period of lecture. No make-up time will be given if you miss some or the whole quiz because of absence or arriving late or leaving early.
 - There will be practice problems posted on the Friday before each quiz.
 - One lowest quiz will be dropped. This is a uniform way of handling illness, family emergency and any other events that may interfere with the course. Thus requests to take quizzes at a time different from the scheduled quiz time will not be granted.
- **Attendance (2%)** This is an in-person class. If you have to miss a lecture, you will need to inform me your absence before the lecture.
 - 2% credit - fewer than two unexcused absences
 - 1% credit - more than two and fewer than four unexcused absences
 - 0% credit - four or more unexcused absence
- **Midterm 1 (20%) - Friday, January 31,** 50-minute exam during class time.
- **Midterm 2 (20%) - Friday, February 28,** 50-minute exam during class time.
- **Final Exam (35%) - Tuesday, March 18 at 10:15am,** 120-minute exam. The final is cumulative.

Quiz and Exam Rules.

- All quizzes and exams are closed-note, closed-book, and without electronic assistance (including calculators and cell phones). Using any notes or electronic device or communicating with anyone except me during an exam constitutes cheating. Cheating on a quiz or exam will be reported to the university with a recommended punishment of failing the course.
- All exams are in person exams in the regular lecture classroom. The attendance of quizzes and exams is mandatory. Requests to take quizzes, midterm exams or the final exam at a time different from the scheduled quiz/exam time will not be granted. Missed midterms will be handled on a case by case basis.
- If you miss the final exam, you will get zero for the final exam. However, if you miss the final exam and are otherwise passing the class, you may request the incomplete grade for the class. See the University rules regarding the eligibility of incomplete grade inquiries under this link <https://provost.uoregon.edu/grades-incompletes-policy>.

Grading Policy. On all written work (e.g. quizzes and exams) submitted for grading, you are expected to show your work clearly and completely; you will be graded on your work as well as on your final answers, but a correct final answer that is neither supported nor justified by work you have shown will receive no credit. I am happy to address any questions regarding the grading of any submitted work, and I encourage you to ask. However, you must bring such questions to my attention within one week of the date it is returned to the class or no changes will be made in the grade. Your final grade of the course is computed according to the following scale:

$$\text{Homework (15\%)} + \text{Attendance (2\%)} + \text{Quiz (8\%)} + \text{Midterm 1 (20\%)} \\ + \text{Midterm 2 (20\%)} + \text{Final (35\%)}$$

Your final percentage will be rounded to the nearest whole number. The letter grade will be assigned based on the following table. Note that the instructor reserves the right to adjust grades at the end of the term based on the outcomes of the course.

Percentage	Grade	Percentage	Grade	Percentage	Grade	Percentage	Grade	Percentage	Grade
98%–100%	A+	88%, 89%	B+	78%, 79%	C+	68%, 69%	D+	0 – 59%	F
92%–97%	A	82%–87%	B	72%–77%	C	62%–67%	D		
90%, 91%	A–	80%, 81%	B–	70%, 71%	C–	60%, 61%	D–		

Note: Any questions regarding grades will not be answered or discussed via email. You will need to meet with me to discuss any concerns or questions about your grades.

Classroom Expectations.

- Students are expected to behave respectfully towards each other and toward the instructor during class time. This includes arriving class on time, and not leaving the class early.
- You may use your tablets or computer to take notes. Otherwise, cell phones, and any electronic devices are not permitted for any activities that are unrelated to the course during lecture time.
- You should read the sections in the textbook and preview the posted slides before they are covered in the class.
- Arrive before class begins and exit only after class is dismissed. It is distracting to have students entering and exiting, so plan accordingly. If you have to miss a class, you are responsible to make up all the materials you miss.

Canvas. Canvas is the place to go for course information and engagement outside of class time. You are responsible for any announcements made on Canvas. To access our course Canvas site, log into <https://canvas.uoregon.edu> using your DuckID. If you have questions about using Canvas, visit the Canvas support page. Canvas and Technology Support also is available by phone (541-346-4357) or by live chat on the Live Help webpage.

Learning Environment. The University of Oregon and I are dedicated to fostering inclusive learning environments for all students and welcomes students with disabilities into all of the University's educational programs. The Accessible Education Center (AEC) assists students with disabilities in reducing campus-wide and classroom-related barriers. If you have or think you have a disability (<https://aec.uoregon.edu/content/what-disability>) and experience academic barriers, please contact the AEC to discuss appropriate accommodations or support. Visit 360 Oregon Hall or <https://aec.uoregon.edu> for more information. You can contact AEC at 541-346-1155 or via email at uoaec@uoregon.edu.

Academic Conduct. All students are expected to maintain academic integrity. Cheating will not be tolerated. Cheating includes but is not limited to use of any unauthorized assistance in taking quizzes, tests, or examinations or knowingly providing any unauthorized assistance to another student on quizzes, tests, or examinations. The University Student Conduct Code (<https://dos.uoregon.edu/code-procedures>) defines academic misconduct. Students are prohibited from committing or attempting to commit any act that constitutes academic misconduct. By way of example, students should not give or receive (or attempt to give or receive) unauthorized help on assignments or examinations without express permission from the instructor. Students should properly acknowledge and document all sources of information (e.g. quotations, paraphrases, ideas) and use only the sources and resources authorized by the instructor. If there is any question about whether an act constitutes academic misconduct, it is the students' obligation to clarify the question with the instructor before committing or attempting to commit the act. Additional information about a common form of academic misconduct, plagiarism, is available at <https://researchguides.uoregon.edu/citing-plagiarism>. All suspected cases of academic misconduct will be reported to the Dean of Students' Office of Student Conduct and Community Standards.

Inclement Weather. It is generally expected that class will meet unless the University is officially closed for inclement weather. If it becomes necessary to cancel class, move the class online (via Zoom), or reschedule an exam while the University remains open, this will be announced on Canvas and by email. Updates on inclement weather and closure are also communicated as described on <https://hr.uoregon.edu/content/inclement-weather-immediate-updates>.

Course Learning Outcomes.

1. Set up and evaluate formulas for Riemann sums, given the function, interval, and number of rectangles.
2. State and use the fundamental theorem of calculus.
3. Evaluate integrals of polynomial and exponential functions, as well as sine and cosine.
4. Evaluate integrals using substitution and integration by parts.
5. Evaluate integrals using appropriate trigonometric substitutions.
6. Use standard trig identities where appropriate as part of integral computations for some trig functions.
7. Interpret the area between two graphs as an integral.
8. Interpret an integral as a signed area.

9. Set up one-variable integrals that represent the solutions to a variety of modeling problems.
10. Evaluate improper integrals.
11. Compute volumes of revolution using both the disk and shell methods, and recognize which method is most appropriate to a given problem.
12. Compute average values of functions over a closed interval.
13. Compute mass, center of mass, and work using integrals.
14. Determine if a given function is a solution to a given differential equation.
15. Write down a differential equation that models a given situation that is described in words, typically where the rate of growth is a linear function of the amount.
16. Find general and particular solutions to basic separable differential equations.

Information about WebWork.

- *Showing Work:* When working on your assignment you should have scratch paper available and neatly write out your thought process in solving the problem. While WebWork does not grade you on this process, writing it out carefully will train you in the skills you need. It will help you track down mistakes, and it will help us track down mistakes when you ask for our help. If you ask us a question about a homework problem in office hours, the first thing we will probably do is ask you to show us your work. Also, remember that on quizzes and exams showing your work will be required. It is important to practice this while doing your homework assignments.
- *Logging into to WebWork:*
 - First go to the main login page at <https://uowebwork.uoregon.edu/webwork2/>.
 - Select the “Math252-23526” section.
 - Your username is your DuckID: for instance, if your uoregon email address is johndoe@uoregon.edu, your DuckID is “johndoe” (without the quotation marks). Your password is the same as your UO email password.
- *Getting Help:*
 - If you have questions about a homework problem, a great way to get assistance specific to your problem is to use the “Email Instructor” button at the bottom of the WebWork. Be sure to type a short message or include your work about what you’ve tried on the problem.
 - You may also send me an email with the original problem along with a description of how you solved the problem as well as any work you did for intermediate steps. Simply asking questions such as “What am I doing wrong on this problem” or “I cannot get the right answer” may not get a proper answer to your question.